

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> To find the current drawn by the head light bulb , we can use the formula : Power (P) = Voltage (V) x Current (I) x Resistance (R). We are given that the power dissipated is 40 watts and the voltage is 12 volts . We can rearrange the formula to solve for current : $I = P / V$. Plugging in the values , we get $I = 40 / 12 = 3 . 33$ amps . Now that we have the current , we can use Oh m 's law to find the resistance : $R = V / I$. Plugging in the values , we get $R = 12 / 3 . 33 = 3 . 6$ oh ms . </think><answer> E . 3 . 33 amps , 3 . 6 oh ms </answer>

